

Mini SIEM Capstone Project – Week 10 Progress Report

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Project Overview

The goal of this project is to design and implement a simplified Security Information and Event Management (SIEM) system capable of ingesting log data, normalizing events into a unified schema, storing data within a database, correlating activity across multiple log sources, generating behavior-based detections, validating alerts using known attacker activity, and visualizing detection results through a web-based dashboard.

During Week 10, the project focused on developing a functional user interface that integrates directly with the SIEM backend and provides live visualization and interaction with generated alerts.

Summary of Week 10 Work

During Week 10, a Flask-based dashboard was developed to serve as the front-end interface for the Mini SIEM platform. The dashboard connects directly to the SQLite backend database and displays live detection metrics, severity classifications, validation statistics, recent alert activity, and interactive filtering functionality.

The dashboard was designed to simulate the workflow of a real SIEM monitoring environment by allowing users to visualize alerts, review detection trends, and filter suspicious activity based on severity, detection rule, and affected system.

This phase completed the transition from backend-only processing to a fully integrated SIEM workflow that includes monitoring, visualization, and interactive analysis capabilities.

Implementation Details

1. Flask Dashboard Development

A web application was developed using the Flask framework to provide a browser-based interface for the SIEM system. The Flask application (`app.py`) retrieves data directly from the SQLite database and dynamically renders the results using HTML templates.

The Flask backend performs the following tasks:

- connects to the SQLite database
- retrieves event and alert metrics
- aggregates alerts by severity and rule type
- calculates validation statistics
- retrieves recent alert activity
- processes dashboard filtering requests
- renders live data within the dashboard interface

The application operates locally and provides real-time visualization of the SIEM detection pipeline.

2. Dashboard Metrics and Visualization

The dashboard includes multiple monitoring sections that summarize overall SIEM activity and detection performance.

Displayed metrics include:

- Total Events Processed
- Total Alerts Generated
- Redteam-Validated Alerts
- Critical Alerts
- Validation Rate

These metrics provide a high-level overview of the SIEM environment and demonstrate the effectiveness of the implemented detection rules.

3. Alert Visualization Charts

Chart.js was integrated into the dashboard to visualize alert distributions and detection trends.

Two primary visualizations were implemented:

- **Alerts by Severity**
Displays alert counts grouped by severity level (critical, high, low).
- **Alerts by Rule**
Displays the distribution of alerts generated by each detection rule.

These visualizations improve readability and allow users to quickly identify dominant alert categories and suspicious activity patterns.

4. Recent Alerts Table

A live alert table was implemented to display recently generated alerts directly from the SQLite database.

Displayed alert information includes:

- timestamp
- rule name
- severity
- affected system
- alert description

This section simulates the alert monitoring functionality commonly found within enterprise SIEM platforms and provides visibility into generated detection events.

5. Interactive Dashboard Filtering

Interactive filtering functionality was added to improve usability and simulate analyst workflows commonly found in modern SIEM environments.

Users can dynamically filter alerts by:

- severity level
- detection rule
- affected system

These filters allow users to quickly isolate specific alert categories and investigate suspicious behavior more efficiently.

Filtering is implemented using Flask request parameters and dynamically updates the alert table without modifying the underlying dataset.

6. Detection Rule Documentation

An additional dashboard section was created to document the implemented detection rules and describe the behavior each rule identifies.

Rules documented include:

- timebound_auth_process_chain
- timebound_process_network_chain
- auth_process_network_chain
- redteam_dns_overlap

This section improves dashboard usability by helping users understand how alerts are generated and what activity patterns are being detected.

7. User Interface Design and Styling

Custom styling was implemented using CSS to improve readability and provide a professional dashboard appearance.

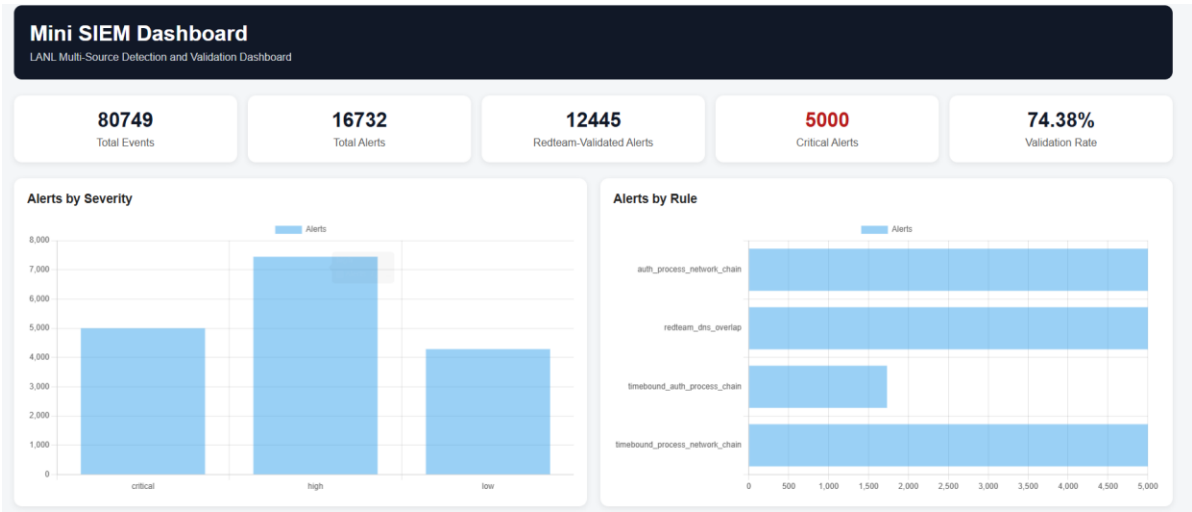
Design improvements include:

- metric summary cards
- responsive dashboard panels
- severity color coding
- chart integration
- styled alert tables
- interactive filtering controls
- dashboard footer and project branding

The final interface provides a clean and organized monitoring environment suitable for demonstrations and presentations.

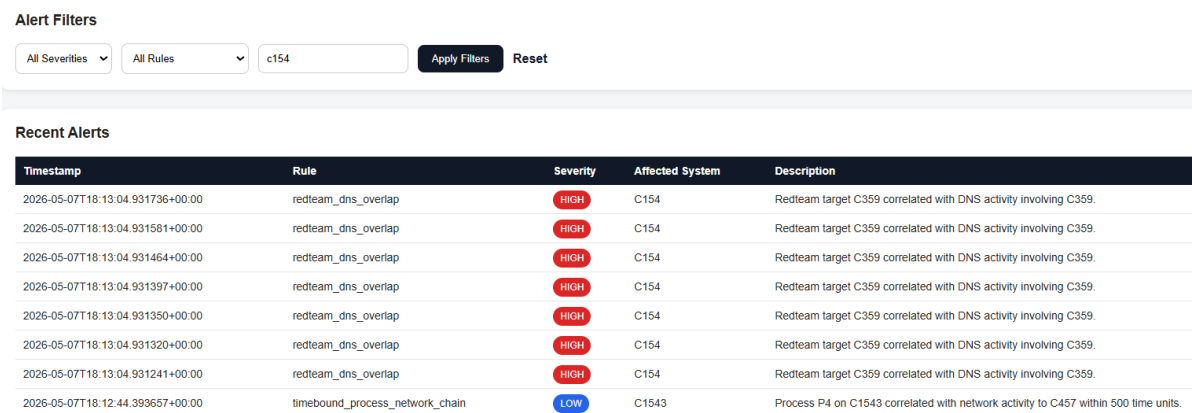
Dashboard Results and Screenshots

Dashboard Overview:



Mini SIEM dashboard displaying event metrics, validation statistics, and alert visualizations.

Interactive Alert Filtering:



Interactive dashboard filtering allowing alerts to be searched and filtered by severity, rule, and affected system.

Detection Rules and Monitoring Interface:

Detection Rules

- **timebound_auth_process_chain** — Authentication followed by process execution within a short time window.
- **timebound_process_network_chain** — Process execution followed by outbound network activity.
- **auth_process_network_chain** — Multi-stage correlation chain involving authentication, process execution, and network communication.
- **redteam_dns_overlap** — DNS activity associated with systems involved in known attacker behavior.

Detection rule documentation section describing implemented behavior-based detection logic.

Key Takeaways

The most significant achievement of Week 10 was the successful integration of a user-facing dashboard with the SIEM backend infrastructure. The project now includes a complete workflow consisting of:

- log ingestion
- normalization
- event storage
- multi-source correlation
- behavior-based detection
- alert validation
- alert persistence
- dashboard visualization
- interactive alert filtering

The dashboard significantly improves usability by allowing detection results to be monitored, filtered, and interpreted visually rather than solely through terminal output.

This phase demonstrates how SIEM systems combine backend analytics with frontend monitoring interfaces to support cybersecurity analysis and incident response workflows.

Challenges Encountered

- Integrating Flask with the SQLite backend
- Aggregating alert data for visualization
- Formatting live alert data for display
- Designing readable chart visualizations
- Implementing dynamic alert filtering and query handling
- Organizing dashboard components for usability
- Maintaining performance while rendering large alert datasets

All challenges were successfully resolved while preserving compatibility with the existing detection engine and database structure.

Final Project Status

At the completion of Week 10, the Mini SIEM project now includes:

- multi-source log ingestion
- unified event normalization
- SQLite event storage
- event correlation engine
- behavior-based detection rules
- time-based correlation logic
- severity classification
- redteam validation
- alert persistence
- Flask-based monitoring dashboard
- interactive alert filtering